

Concept Questions: Chapter 1

Dr. Kyle Horne



ME-3640 – Heat Transfer

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Abstract

Fundamental principals and concepts are far more important to engineering expertise than understanding of example systems. These concept questions (shamelessly lifted from the textbook) will help you test your knowledge after reading the chapter summary and building a concept map. Use the relative ease and difficulty you experience while answering these questions to prioritize your efforts fo the remainder of this chapter.

1 Thermodynamics and Heat Transfer

- 1-1C** How does the science of heat transfer differ from the science of thermodynamics?
- 1-2C** What is the driving force for (a) heat transfer, (b) electric current flow, and (c) fluid flow?
- 1-3C** What is the caloric theory? When and why was it abandoned?
- 1-4C** How do rating problems in heat transfer differ from the sizing problems?
- 1-5C** What is the difference between the analytical and experimental approach to heat transfer? Discuss the advantages and disadvantages of each approach.
- 1-6C** How do the differential equations in the study of a physical problem arise?
- 1-7C** What is the importance of modeling in engineering? How are the mathematical models for engineering processes prepared?
- 1-8C** When modeling an engineering process, how is the right choice made between a simple but crude and a complex but accurate model? Is the complex model necessarily a better choice since it is more accurate?
- 1-9C** On a hot summer day, a student turns his fan on when he leaves his room in the morning. When he returns in the evening, will his room be warmer or cooler than the neighboring rooms? Why? Assume all the doors and windows are kept closed.
- 1-10C** Consider two identical rooms, one with a refrigerator in it and the other without one. If all the doors and windows are closed, will the room that contains the refrigerator be cooler or warmer than the other room? Why?
- 1-11C** An ideal gas is heated from 50°C to 80°C (a) at constant volume and (b) at constant pressure. For which case do you think the energy required will be greater? Why?
- 1-12C** How are heat, internal energy, and thermal energy related to each other?
- 1-13C** What is heat flux? How is it related to the heat transfer rate?
- 1-14C** What are the mechanisms of energy transfer to a closed system? How is heat transfer distinguished from the other forms of energy transfer?

2 Heat Transfer Mechanisms

- 1-33C** Define thermal conductivity and explain its significance in heat transfer.
- 1-34C** Judging from its unit $\text{W/m}\cdot\text{K}$, can we define thermal conductivity of a material as the rate of heat transfer through the material per unit thickness per unit temperature difference? Explain.
- 1-35C** Which is a better heat conductor, diamond or silver?
- 1-36C** How do the thermal conductivity of gases and liquids vary with temperature?
- 1-37C** Why is the thermal conductivity of superinsulation orders of magnitude lower than the thermal conductivity of ordinary insulation?
- 1-38C** Why do we characterize the heat conduction ability of insulators in terms of their apparent thermal conductivity instead of the ordinary thermal conductivity?
- 1-39C** Consider an alloy of two metals whose thermal conductivities are k_1 and k_2 . Will the thermal conductivity of the alloy be less than k_1 , greater than k_2 , or between k_1 and k_2 ?
- 1-40C** What are the mechanisms of heat transfer? How are they distinguished from each other?
- 1-41C** Write down the expressions for the physical laws that govern each mode of heat transfer, and identify the variables involved in each relation.
- 1-42C** How does heat conduction differ from convection?
- 1-43C** Does any of the energy of the sun reach the earth by conduction or convection?
- 1-44C** How does forced convection differ from natural convection?
- 1-45C** What is the physical mechanism of heat conduction in a solid, a liquid, and a gas?
- 1-46C** Consider heat transfer through a windowless wall of a house on a winter day. Discuss the parameters that affect the rate of heat conduction through the wall.
- 1-47C** Consider heat loss through the two walls of a house on a winter night. The walls are identical, except that one of them has a tightly fit glass window. Through which wall will the house lose more heat? Explain.
- 1-48C** Consider two houses that are identical, except that the walls are built using bricks in one house, and wood in the other. If the walls of the brick house are twice as thick, which house do you think will be more energy efficient?
- 1-49C** Consider two walls of a house that are identical except that one is made of 10-cm-thick wood, while the other is made of 25-cm-thick brick. Through which wall will the house lose more heat in winter?
- 1-50C** Define emissivity and absorptivity. What is Kirchhoff's law of radiation?
- 1-51C** What is a blackbody? How do real bodies differ from blackbodies?

3 Simultaneous Heat Transfer Mechanisms

1-90C Can all three modes of heat transfer occur simultaneously (in parallel) in a medium?

1-91C Can a medium involve (a) conduction and convection, (b) conduction and radiation, or (c) convection and radiation simultaneously? Give examples for the “yes” answers.

1-92C The deep human body temperature of a healthy person remains constant at 37°C while the temperature and the humidity of the environment change with time. Discuss the heat transfer mechanisms between the human body and the environment both in summer and winter, and explain how a person can keep cooler in summer and warmer in winter.

1-93C We often turn the fan on in summer to help us cool. Explain how a fan makes us feel cooler in the summer. Also explain why some people use ceiling fans also in winter.

A Rubrics

Submissions of this assignment will be graded upon the criteria found in this section. Review these carefully, and complete your work so that application of this rubric produces your desired score.

 Criterion 1 Problems	
Homework is completed with appropriate effort, neatness, etc.	
Points	Description
20	Honest effort to participate and learn: every problem shows an answer or questions which need resolution.
10	Rushed or haphazard work: some problems show no effort at all.
0	Work is absent, wholly incorrect, or plagiarized.